



Student Name: _____

Teacher Code: HEWP THOS LOPF TRIA AJIA CHEY

Test Name: SPECIALIST MATHEMATICS

MATHEMATICS TEST

UNIT 2

Test theme: EXAM TECH FREE

Reading time: 15 minutes

Writing time: 60 minutes

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
10	10	50

- Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers.
- Students are NOT permitted to bring into the examination: notes of any kind, blank sheets of paper, white out liquid/tape or a calculator of any type.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your name in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

- Answer all questions in the spaces provided.
- Unless otherwise specified, an exact answer is required.
- In questions where more than one mark is available, appropriate working must be shown.
- Diagrams are not necessarily drawn to scale.

Question 1**(5 marks)**

Consider the function

$$f : D \rightarrow \mathbb{R}, \quad f(x) = \frac{1}{2 \sin^2 x - \sin x}.$$

a. State the implied domain of $f(x)$ for $x \in [0, \pi]$.**1 mark****b.** Sketch the graph of $y = f(x)$ for $x \in [0, \pi]$. Label all vertical asymptotes with equations and local extrema with exact coordinates.**4 marks**

Question 2**(4 marks)**

Let

$$A = \begin{pmatrix} k-1 & 2 \\ 4 & k+1 \end{pmatrix}.$$

a. Determine the value(s) of k for which A is singular.**1 mark****b.** For $k = 4$, solve the matrix equation

$$AX = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \end{pmatrix},$$

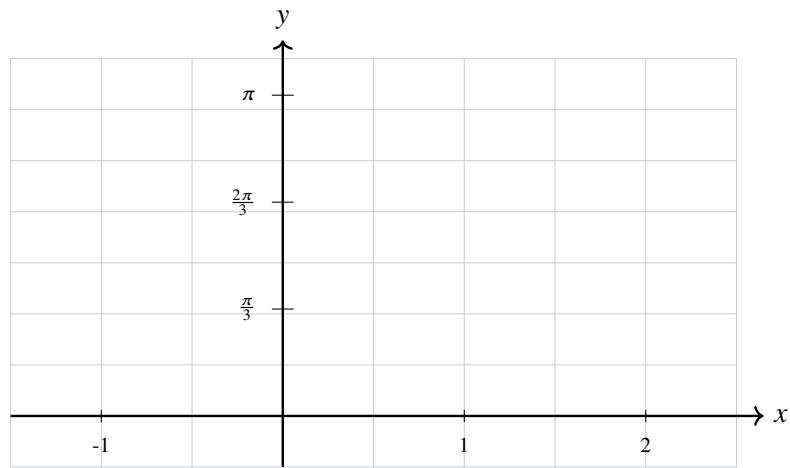
using the matrix inverse.

3 marks**Question 3****(5 marks)**

Consider the function

$$h(x) = \arcsin\left(\frac{2x-1}{3}\right) + \frac{2\pi}{3}.$$

a. State the domain and range of $h(x)$.**2 marks****b.** Sketch the graph of $h(x)$, labelling endpoints and intercepts with exact coordinates.**3 marks**

**Question 4****(5 marks)**

- a. Find all complex roots of the equation

$$z^4 + 2z^2 + 4 = 0.$$

Give your answers in Cartesian form.

3 marks

- b. Show that these roots lie on a circle centred at the origin in the Argand diagram and state the radius. **2 marks**

Question 5**(6 marks)****a.** Sketch the subset

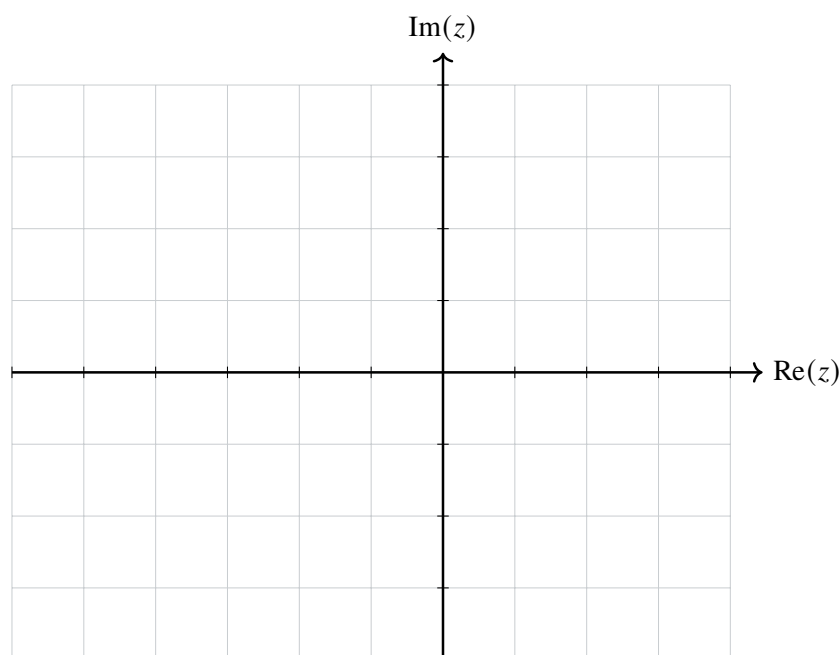
$$S_1 = \{z \in \mathbb{C} : |z - 2| = 2|z + 1|\}.$$

Show algebraically that this is a circle and label its centre and radius.

3 marks

b. On the same diagram, sketch

$$S_2 = \{z \in \mathbb{C} : \operatorname{Re}(z) \leq -2\} \cup \left\{z \in \mathbb{C} : \frac{3\pi}{4} \leq \operatorname{Arg}(z + 3) \leq \pi\right\}.$$

3 marks

Question 6**(4 marks)**

- a.** Let $n \in \mathbb{Z}$. Use proof by contrapositive to show that if n^2 is a multiple of 3, then n is a multiple of 3. **2 marks**

- b.** Use the result from part **a** to prove by contradiction that $\sqrt{3}$ is irrational. **2 marks**

Question 7**(5 marks)**

A particle moves in a straight line with velocity

$$v(t) = t^3 - 6t^2 + 8t, \quad t \geq 0.$$

- a.** Find the times t when the particle is at rest. **1 mark**

- b.** Calculate the total distance travelled by the particle in the first 4 seconds of motion. **4 marks**

Question 8**(4 marks)****a.** Express

$$\cos x - \sqrt{3} \sin x$$

in the form $R \cos(x + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.**2 marks****b.** Hence, find the exact value of

$$\cos\left(\frac{\pi}{12}\right) - \sqrt{3} \sin\left(\frac{\pi}{12}\right).$$

2 marks**Question 9****(6 marks)**Prove by mathematical induction that for all $n \in \mathbb{Z}^+$,

$$\sum_{r=1}^n \frac{r}{2^r} = 2 - \frac{n+2}{2^n}.$$

6 marks

Question 10**(6 marks)**

Consider the equation $z^n = 1$ for $n = 6$. The roots are denoted

$$\omega_k = \operatorname{cis}\left(\frac{2\pi k}{6}\right), \quad k = 0, 1, \dots, 5.$$

a. Show that

$$\omega_1 + \omega_2 + \omega_3 + \omega_4 + \omega_5 = -1.$$

1 mark

b. Let P be the point in the Argand diagram representing the complex number $z = 1$. Let A, B, C, D, E be the points representing the other five roots.

Find the exact value of

$$|PA| |PB| |PC| |PD| |PE|.$$

5 marks**END OF EXAM**